

## Lab 3: Component Modelling & Architectural Pattern Selection

**Problem Statement: Disaster Relief Supply & Volunteer Coordinator**

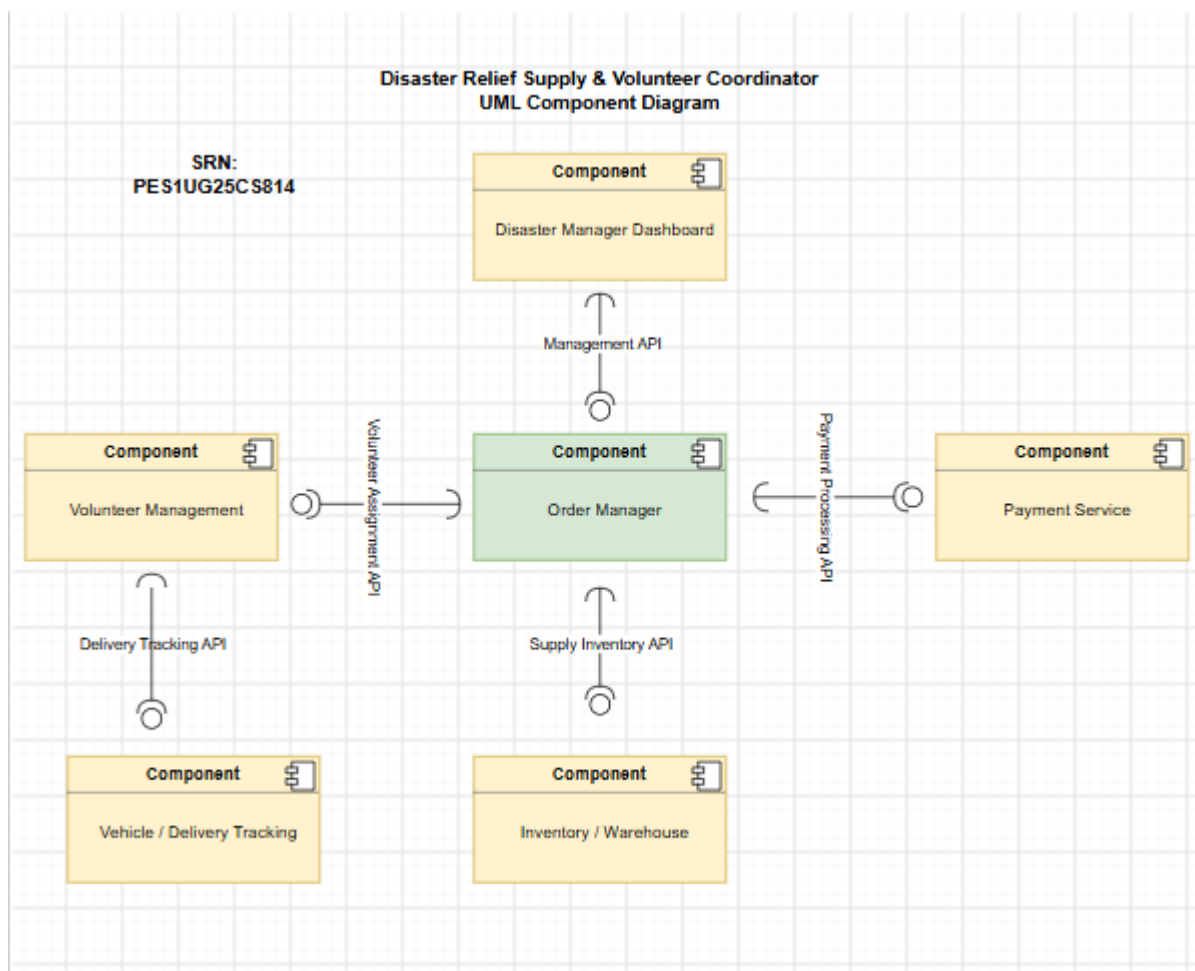
**Architecture: Microservices Architecture**

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**Step:1**



### Architecture Selection

We chose **Microservices Architecture** for the Disaster Relief Supply & Volunteer Coordinator System. The system is divided into independent services such as Order Management, Payment Service, Inventory/Warehouse, Volunteer Management, Vehicle/Delivery Tracking, and the Disaster Manager Dashboard.

### **Reason 1 – Independent Scaling**

During a disaster, the number of shelter supply requests and delivery updates can increase significantly. Microservices allow heavily used services such as Order Manager and Inventory/Warehouse to be scaled independently without scaling the entire system.

### **Reason 2 – Fault Isolation**

The system contains separate functions for supply management, volunteer coordination, payment processing, and vehicle tracking. If one service experiences a failure, the other services can continue operating independently, improving the resilience of the disaster-relief platform.

### **Security Advantage**

Microservices provide clear service boundaries, allowing access to individual services through controlled interfaces. Authentication and authorization can therefore be applied separately to sensitive operations such as payment processing and Disaster Manager functions.

### **Performance Benefit**

Frequently used services such as supply and order management can be scaled independently. This helps the system handle a large number of shelter requests during peak disaster situations without unnecessarily scaling less-used services.